

ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

Objective

Configure two virtual machines to communicate with each other over an isolated internal network, with no access to the host machine or the internet.

Overview

By default, VirtualBox assigns virtual machines a NAT adapter, which provides internet access but prevents VMs from communicating with each other. To enable VM-to-VM communication without involving the host machine, both adapters are switched to Internal Network mode and assigned static IP addresses on the same subnet.

Environment

Machine	OS	Role	IP Address
VM 1	Windows 11	Node A	192.168.1.3
VM 2	Ubuntu	Node B	192.168.1.4

Step-by-Step Instructions

Step 1: Configure the Network Adapter on VM 1 (Windows 11)

1. Open VirtualBox Manager.
2. Right-click VM 1 and select Settings.
3. Navigate to Network and select Adapter 1.
4. Ensure Enable Network Adapter is checked.
5. Set Attached to: Internal Network.
6. Set the Name field to: TCM (this name must match exactly on both VMs).
7. Click OK.

Step 2: Configure the Network Adapter on VM 2 (Ubuntu)

Repeat the same steps for VM 2:

1. Right-click VM 2 and select Settings.
2. Navigate to Network and select Adapter 1.
3. Set Attached to: Internal Network.
4. Set the Name field to: TCM (must be identical to the name used on VM 1).
5. Click OK.

Important: Both VMs must use the exact same Internal Network name. A typo will create two separate, isolated networks, and the VMs will not be able to communicate with each other.

ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

Step 3: Assign Static IP Addresses

With both VMs running, static IP addresses must be assigned manually. Because there is no DHCP server on an Internal Network, the addresses cannot be assigned automatically.

On VM 1 (Windows 11), open a Linux-style shell or WSL and run:

```
sudo ip addr add 192.168.1.3/24 dev eth0
sudo ip link set eth0 up
```

On VM 2 (Ubuntu), run:

```
sudo ip addr add 192.168.1.4/24 dev enp0s3
sudo ip link set enp0s3 up
```

Note: These commands apply only for the current session. The settings will be lost after a reboot. See Step 4 for a permanent configuration.

Step 4: Make the IP Address Persistent on Ubuntu (VM 2)

To retain the IP address after a reboot, configure Netplan by opening the configuration file:

```
sudo nano /etc/netplan/01-netcfg.yaml
```

Enter the following configuration:

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.4/24
```

Apply the configuration:

```
sudo netplan apply
```

Step 5: Test the Connection

From VM 1, ping VM 2:

```
ping 192.168.1.4
```

ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

From VM 2, ping VM 1:

```
ping 192.168.1.3
```

A successful ping response confirms that the two VMs are communicating over the internal network.

Note: Pinging 127.0.0.1 only tests the local loopback interface on the machine where the command is run. It does not verify connectivity between two separate VMs. To confirm VM-to-VM communication, always ping the other machine's IP address as shown above.

How It Works

The Internal Network adapter type creates a private virtual switch that only the VMs assigned to the same named network can join. Traffic never passes through the host machine's network stack, which is why the host remains unreachable from either VM.

Key Concepts

Internal Network: A VirtualBox network mode that creates an isolated switch for VM-to-VM traffic only.

Static IP: A manually assigned IP address which is required because Internal Network mode has no DHCP server.

Subnet /24: Covers addresses from 192.168.1.0 to 192.168.1.255. Both VMs must be assigned addresses within this range.

Netplan: Ubuntu's built-in network configuration tool that reads YAML files and applies network settings persistently.

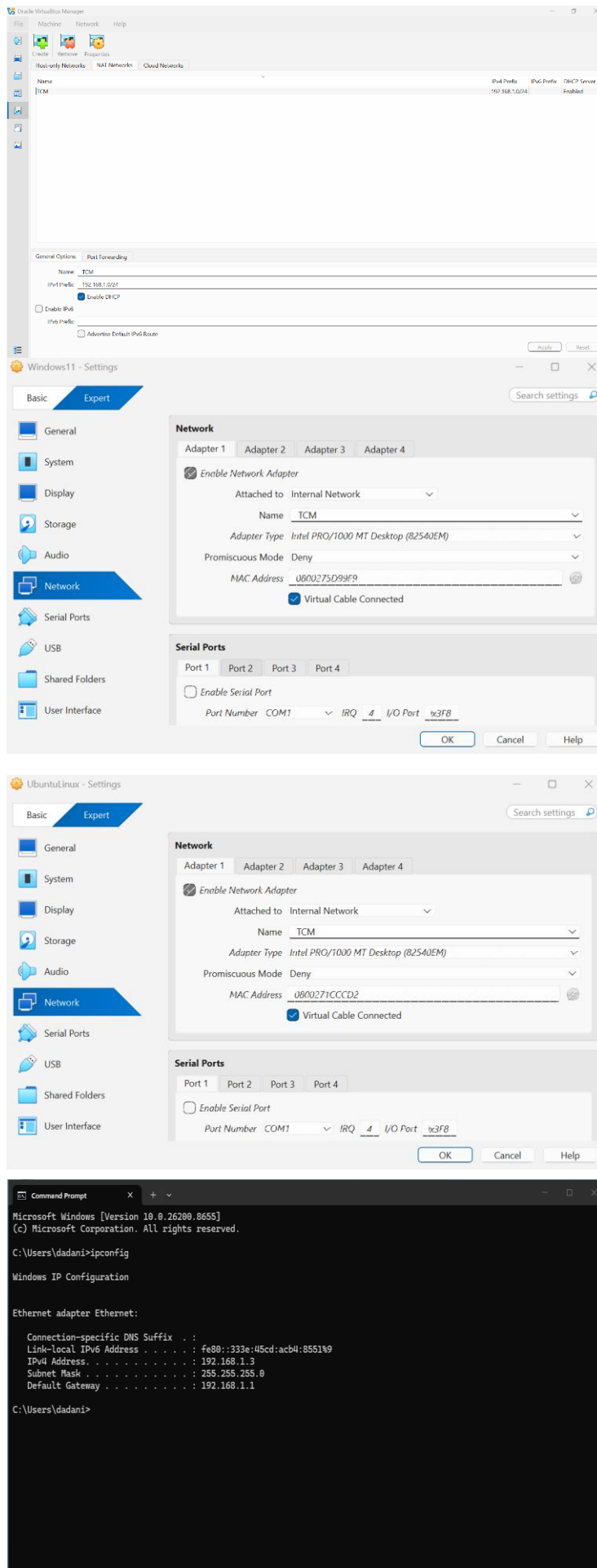
Loopback (127.0.0.1): A local address that tests only the machine's own network stack. It cannot be used to test connectivity between two machines.

References

- VirtualBox Networking Modes: <https://www.virtualbox.org/manual/ch06.html>
- Ubuntu Netplan Documentation: <https://netplan.io/>

ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

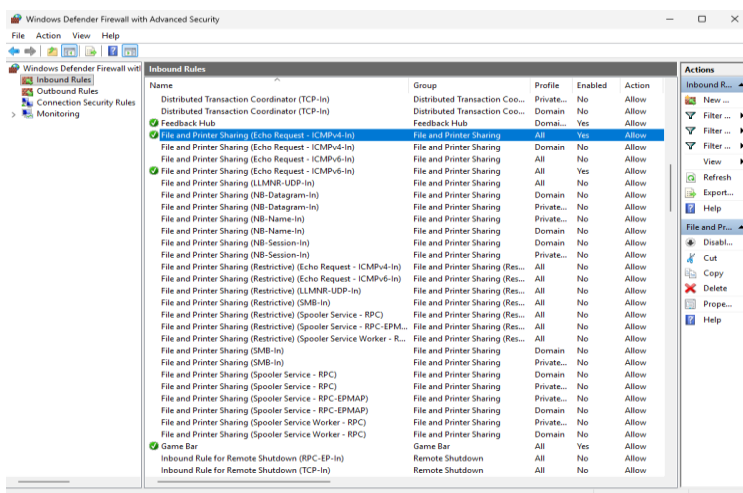
Screenshots



ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

```
dadani@UbuntuLinux: ~  
dadani@UbuntuLinux:~$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.1.4 netmask 255.255.255.0 broadcast 192.168.1.255  
    inet6 fe80::a00:27ff:fe1c:ccd2 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:1c:cc:d2 txqueuelen 1000 (Ethernet)  
    RX packets 1876 bytes 270650 (270.6 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 178 bytes 20701 (20.7 KB)  
    TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1980 bytes 149386 (149.3 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1980 bytes 149386 (149.3 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
dadani@UbuntuLinux:~$
```

```
Microsoft Windows [Version 10.0.26200.855]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\dadani>ipconfig  
  
Windows IP Configuration  
  
Ethernet adapter Ethernet:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::333e:45cd:acb4:8551%9  
    IPv4 Address. . . . . : 192.168.1.3  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 192.168.1.1  
  
C:\Users\dadani>ping 192.168.1.4  
  
Pinging 192.168.1.4 with 32 bytes of data:  
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64  
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64  
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64  
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64  
  
Ping statistics for 192.168.1.4:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 1ms, Average = 0ms  
  
C:\Users\dadani>
```



ITF Lab: VM-to-VM Communication Using VirtualBox Internal Network

```
dadani@UbuntuLinux: ~ — ping 192.168.1.3
dadani@UbuntuLinux:~$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data:
64 bytes from 192.168.1.3: icmp_seq=1 ttl=128 time=1.22 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=128 time=1.11 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=128 time=1.02 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=128 time=1.06 ms
64 bytes from 192.168.1.3: icmp_seq=5 ttl=128 time=1.37 ms
64 bytes from 192.168.1.3: icmp_seq=6 ttl=128 time=1.30 ms
64 bytes from 192.168.1.3: icmp_seq=7 ttl=128 time=0.686 ms
64 bytes from 192.168.1.3: icmp_seq=8 ttl=128 time=1.02 ms
64 bytes from 192.168.1.3: icmp_seq=9 ttl=128 time=1.50 ms
64 bytes from 192.168.1.3: icmp_seq=10 ttl=128 time=1.69 ms
64 bytes from 192.168.1.3: icmp_seq=11 ttl=128 time=4.28 ms
64 bytes from 192.168.1.3: icmp_seq=12 ttl=128 time=1.19 ms
64 bytes from 192.168.1.3: icmp_seq=13 ttl=128 time=0.817 ms
64 bytes from 192.168.1.3: icmp_seq=14 ttl=128 time=2.25 ms
64 bytes from 192.168.1.3: icmp_seq=15 ttl=128 time=1.47 ms
^
```